Week 4: Databricks ETL for Smart Energy Monitoring

Objective

Build an ETL (Extract, Transform, Load) pipeline in Azure Databricks to process smart home energy logs and generate daily/weekly energy summaries.

Capstone Tasks

- Upload cleaned energy logs to Databricks
- Build an ETL pipeline to calculate daily and weekly summaries
- Save final results in Delta format or CSV
- Optional: use SQL to identify energy-saving opportunities

Tools Used

- Azure Databricks
- PySpark
- Delta Lake

Input File

energy_usage.csv

device_id,room_id,timestamp,energy_kwh

1,101,2026-05-13 10:00:00,2.5

2,102,2026-05-13 10:10:00,5.2

3,103,2026-05-13 10:20:00,1.1

1,101,2026-05-14 11:00:00,3.0

2,102,2026-05-14 11:10:00,4.8

3,103,2026-05-14 11:20:00,2.0

ETL Steps

Step 1: Upload Data to Databricks

Upload energy_usage.csv to Databricks FileStore or workspace storage.

Example path:
/FileStore/tables/energy_usage.csv

Step 2: Load Data Using PySpark

```
from pyspark.sql.functions import col, sum, to_date, weekofyear


# Load CSV
logs = spark.read.csv(
    "/FileStore/tables/energy_usage.csv",
    header=True,
    inferSchema=True
)


logs.show()
```

Step 3: Transform Data

Extract day and week values for aggregation.

```
logs = logs.withColumn("day", to_date(col("timestamp")))
logs = logs.withColumn("week", weekofyear(col("timestamp")))
```

Step 4: Create Daily Summary

```
daily_summary = logs.groupBy("device_id", "day") \
    .agg(sum("energy_kwh").alias("daily_energy"))


daily_summary.show()
```

Step 5: Create Weekly Summary

```
weekly_summary = logs.groupBy("device_id", "week") \
    .agg(sum("energy_kwh").alias("weekly_energy"))


weekly_summary.show()
```

Step 6: Save Final Results

Save as Delta format:

```
daily_summary.write.format("delta").mode("overwrite") \
```

```
    .save("/FileStore/final_energy_delta")
```

Or save as CSV:

```
weekly_summary.write.mode("overwrite") \
    .csv("/FileStore/final_energy_csv", header=True)
```

Step 7 (Optional): Query Energy Savings Opportunities Using SQL

```
SELECT device_id, SUM(energy_kwh) AS total_energy
FROM energy_logs
GROUP BY device_id
ORDER BY total_energy DESC;
```

Sample Output

Daily Summary

device_id	day	daily_energy
1	2026-05-13	2.5
1	2026-05-14	3.0
2	2026-05-13	5.2
2	2026-05-14	4.8

Weekly Summary

device_id	week	weekly_energy
1	20	5.5
2	20	10.0
3	20	3.1

Deliverables

- Databricks notebook with full ETL pipeline
- Output files saved in Delta/CSV format
- Optional SQL query for energy optimization insights